PASS

Modernizing SQL Server the Right Way

Gianluca Hotz, Danilo Dominici, UGISS.ORG

Why upgrade?

Winter is coming…

When will end of support happen?

SQL Server 2008 and 2008 R2 will no longer be supported starting on **July 9, 2019**.

	Current support level	End mainstream	End extended
SQL Server 2014	Currently supporting all versions	July 9, 2019	July 9, 2024
SQL Server 2012	SQL Server 2012 SP2+ is in extended support which includes security updates, paid support, and requires purchasing non-security hotfix support	July 11, 2017	July 12, 2022
⚠ SQL Server 2008 and SQL Server 2008 R2	SQL Server 2008 and 2008 R2 are in extended support which includes security updates, paid support, and requires purchasing non-security hotfix support	July 8, 2014	July 9, 2019
⚠ SQL Server 2005	SQL Server 2005 support ended on April 12, 2016	April 12, 2011	April 12, 2016

Learn more about the SQL Server support lifecycle: support.microsoft.com/lifecycle/

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Is it the right time to modernize?

- Do I need to modernize to reduce costs?
- To maintain compliance?
- To keep vendor support?
- For ISVs, to enable choice and features for my customers?
- Can I separate application modernization cycles from data modernization cycles?

Database Compatibility Certification

Database Compatibility Level based certification

Stop certifying for any given platform (Cloud, on-prem)!

Stop certifying for a named SQL Server version!

Any certification process should be thought in terms of
"which target database compatibility level am I certifying to?"

Updated public documentation: http://aka.ms/dbcompat

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Key Benefits

Simplified application certification on-premise and Azure (e.g. Azure SQL DB MI).

Ability to provide customer a choice of latest SQL Server platform based on certified DB compat level.

Improved risk management by decoupling application upgrade cycles from Database upgrade cycles.

Microsoft stands by DB Compat based certification

Microsoft Database Compatibility Level Protection

Full Functional protection once assessment tools runs clean with no errors.

Query Plan shape protection on comparable hardware.

Maintaining backward compatibility is very important to SQL Server team.

Database Compatibility Level behavior

Database Compatibility Level sets certain database behaviors to be compatible with the specified version of SQL Server.

Compatibility level affects behaviors only for the specified database, not for the entire server.

Product	Compatibility Level Designation	Supported Compatibility Level Values
SQL Server 2019	150	150, 140, 130, 120, 110, 100
SQL Server 2017	140	140, 130, 120, 110, 100
Azure SQL Database	130	150, 140, 130, 120, 110, 100
SQL Server 2016	130	130, 120, 110, 100
SQL Server 2014	120	120, 110, 100
SQL Server 2012	110	110, 100, 90
SQL Server 2008 R2	100	100, 90, 80
SQL Server 2008	100	100, 90, 80
SQL Server 2005	90	90, 80
SQL Server 2000	80	80

Functional change protection Clarifying the caveats

Deprecated = avoid use in new development

- Deprecated functionality introduced in a given SQL Server version is still protected by that compatibility level.

Discontinued = removed from product

- Discontinued functionality introduced in a given SQL Server version is not protected by compatibility level.

Example of removed T-SQL syntax.

- In SQL Server 2012 the `fastfirstrow` hint was removed.
- Regardless of the compatibility level, the query below will produce error 321 (not a recognized table hints option):
  ```
  SELECT * FROM HumanResources.Employee WITH (FASTFIRSTROW);
  ```
- Instead use:
  ```
  SELECT * FROM HumanResources.Employee OPTION (FAST = <n>);
  ```

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Functional change protection Clarifying the caveats

Breaking Changes = behavior changes resulting in different outcome

Protected by Database Compatibility:

```
DECLARE @value datetime = '1900-01-01 00:00:00.003'
SELECT CAST(@value AS datetime2)
```

- In DB Compat 120 or lower, result is:
  `1900-01-01 00:00:00.0030000`
- Under DB Compat 130, these show improved accuracy by accounting for the fractional milliseconds, resulting in:
  `1900-01-01 00:00:00.0033333`

Not Protected by Database Compatibility:

- The query below works until DB Compat 90, but errors out starting with Database Compatibility 100 (error 241, conversion fail):
  `SELECT DATEPART (year, '2007/05-30')`
- Instead use:
  `SELECT DATEPART (year, '2007/05/30')` or `SELECT DATEPART (year, '2007-05-30')`

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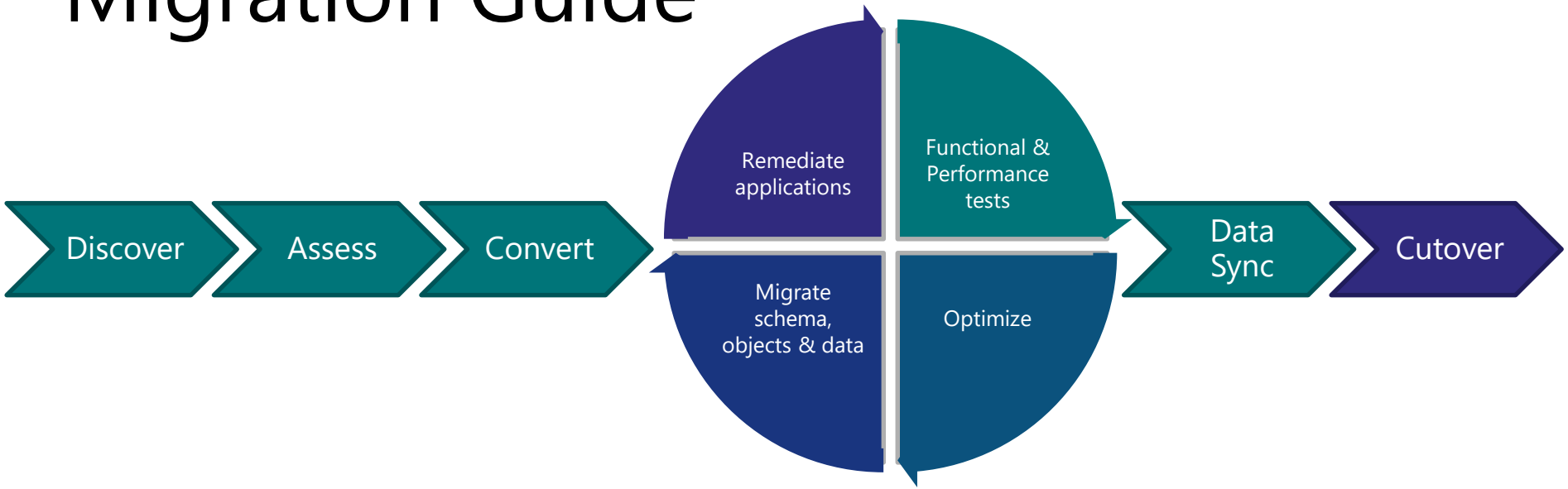
Upgrade Process

Minimize Risk with the Database Migration Guide



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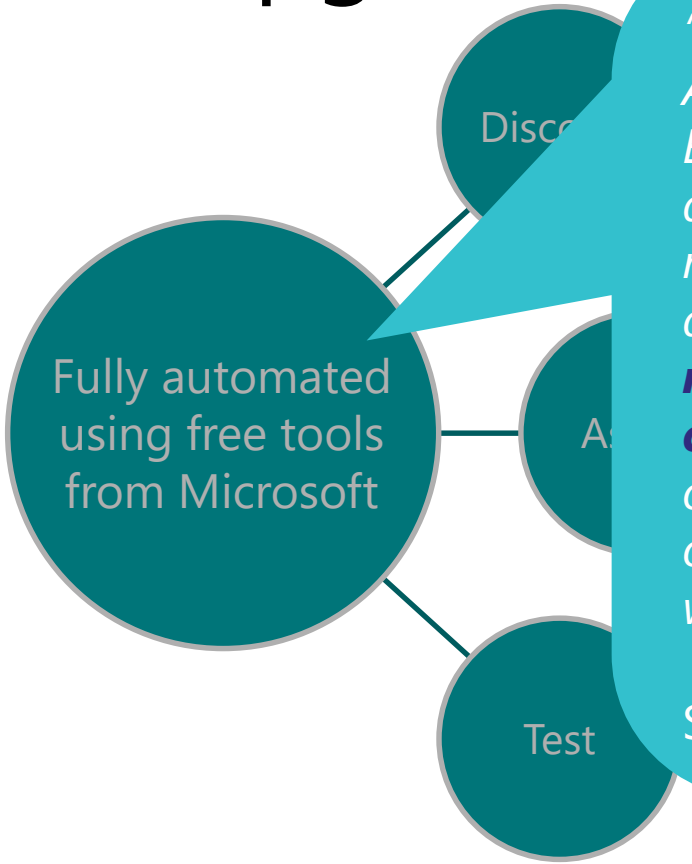
Reliable Upgrades



*"With tools like Database Migration Assistant and Database Experimentation Assistant, we were able to **reduce the time and effort** required for the upgrade, enable automated A/B testing capability to **minimize risk** and provide a **high confidence** upgrade plan for a mission critical, Tier-1 environment spanning over a 1000 instances of SQL Server within 3 months."*

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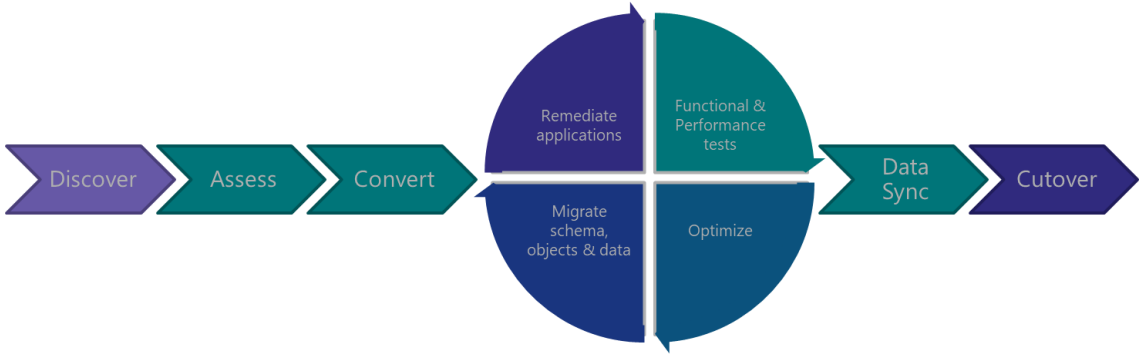
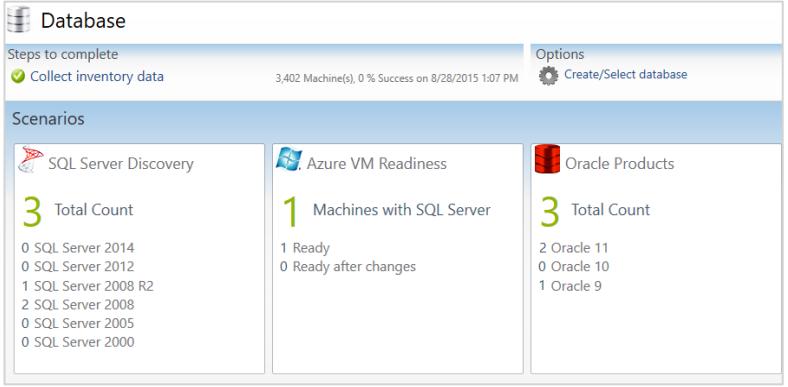
Discover

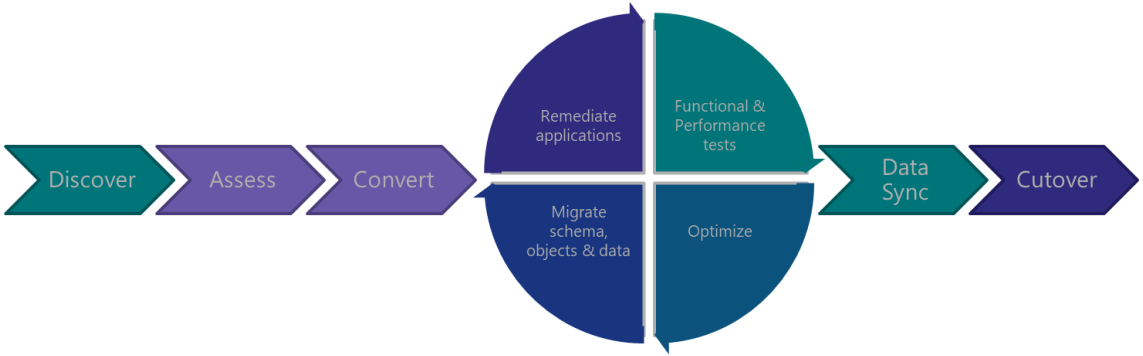

Discover
Assess
Convert
Remediate applications
Functional & Performance tests
Migrate schema, objects & data
Optimize
Data Sync
Cutover

Discover with MAP Toolkit



- Which SQL Server versions do I have?
- Which Editions am I running?
- Which SQL Server components are installed?
- How many cores are on each server?
- How many databases are in each instance?
- What are the sizes of all my databases?
- What are the settings for each instance and database?

Assess & Convert



Assess & Convert with DMA

Data Migration Assistant

*"Upgrade readiness assessment was delivered with the help of **Database Migration Assistant** which helped us analyze our template databases to **identify potential pitfalls** and suggested **automated remediation** for our development teams to act on."*

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Perform Migration

- Connect to source and target servers
- Choose databases
- Monitor migration
- Review final report

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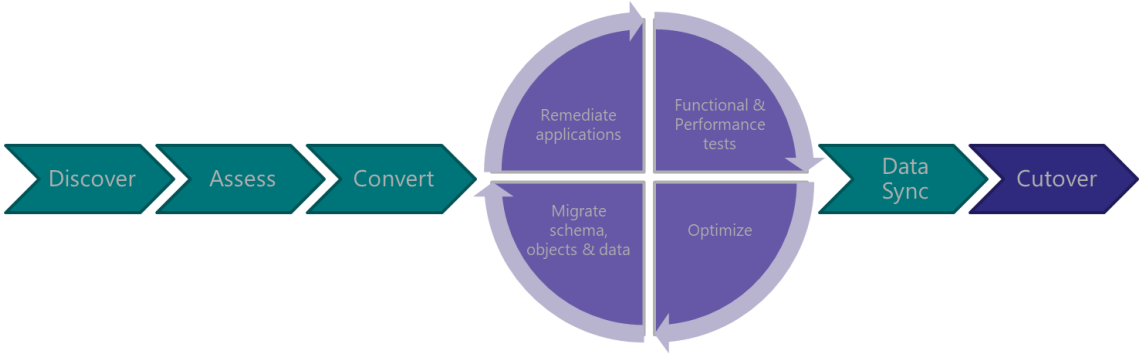
Test & Optimize



Test & Optimize with DEA

*"**Database Experimentation Assistant** helped us perform comparison tests between our current environment and a test SQL Server 2016 environment to identify **regressions**, **breaking changes** and **performance characteristics**.*

*Because of the **automation**, we were able to repeat such an exercise with multiple iterations using **different configuration settings**."*

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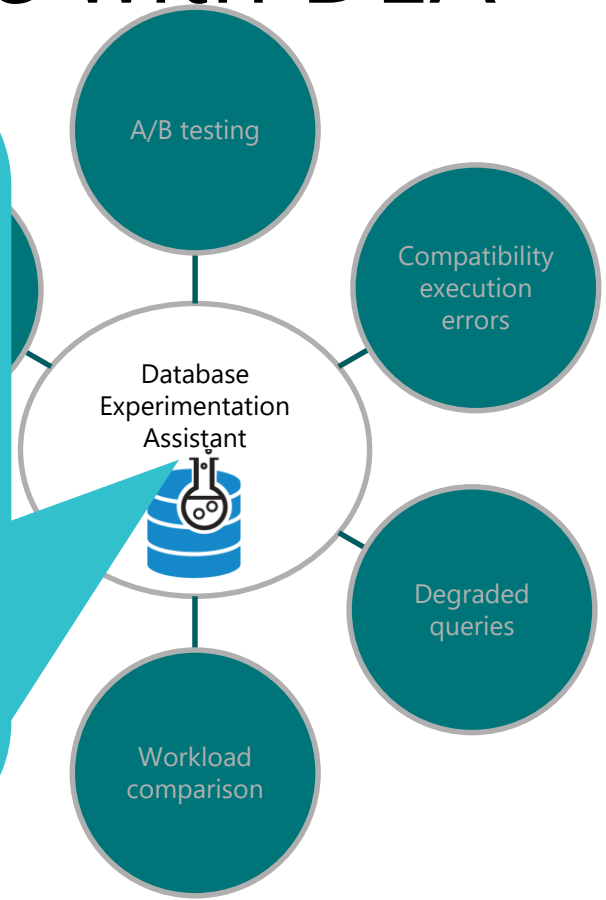

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Walkthrough

Database Migration Assistant

Data Migration Assistant

Inventory-Assessment1

Delete Assessment

1 Options | 2 Select sources | 3 Review results

Select target version

SQL Server 2017 on Windows

Select report type

- [x] **Compatibility Issues**
  Discover breaking changes, behavior changes, and deprecated features by analyzing the databases you chose in your source server to be migrated to a new SQL Server platform.
- [x] **New features' recommendation**
  Discover new SQL Server features that are applicable to the databases and tables in your source server once migrated to the new target SQL Server platform.
- [ ] Check feature parity
  Discover unsupported or partially-supported features and functions that your applications may rely on. Get guidance around these areas that may need some re-engineering.

Next

11:03 PM
7/19/2018

Data Migration Assistant

Inventory-Assessment1

1 Options ✓ | 2 Select sources | 3 Review results

Add sources | Remove sources

Name

Connect to a server

Connect to a server and select sources

Server name

TIGERSQL2008R2

Authentication type

Windows Authentication

Connection properties

- [x] Encrypt connection
- [x] Trust server certificate

SQL Server permissions

To run the selected advisor(s), credentials used to connect to source SQL Server instance must have CONNECT SQL, VIEW SERVER STATE, and VIEW ANY DEFINITION permissions.

Connect

10:58 PM
7/19/2018

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Data Migration Assistant

Inventory-Assessment1

Restart Assessment | Delete Assessment

1 Options ✓ | 2 Select sources ✓ | 3 Review results

- Compatibility issues
- Feature recommendations

Target Platform

SQL Server 2017 on Windows

TIGERSQL2008R2 (SQL Server 2008 R2) (1)

- Inventory

Inventory / SQL Server 2008 R2 Compat 80 Size 5.29 GB

Compatibility 140 (14) | Compatibility 130 (14) | Compatibility 120 (13) | Compatibility 110 (13) | Compatibility 100 (10)

Issue	Impacted objects
Breaking changes (8)	
Discontinued DBCC command...	4
New column in output of 'sp_h...	4
Remove user-defined type (UD...	1
Constant expressions are not a...	1
SQL Mail has been discontinued	1
Detected statements that refer...	1
FOR BROWSE is not allowed in...	1
Table hints in indexed view def...	1
Behavior changes (5)	
SERVERPROPERTY('LCID') resul...	2
Unqualified Join(s) detected	2
FOR XML AUTO queries return...	1
SET ROWCOUNT used in the c...	1
ORDER BY specifies integer or...	1
Deprecated features (1)	
Deprecated data types TEXT, I...	4

Discontinued DBCC commands referenced in your T-SQL objects

Issue details

Impact

Many DBCC commands that were available in prior releases have been replaced with DMVs and DMFs, or no longer exist; therefore, using these commands may cause errors and unforeseen effects after upgrading your SQL Server.

Recommendation

Re-write the code, replace "DBCC DBREINDEX" with "ALTER INDEX" with "REBUILD" option.

Re-write the code, replace "DBCC INDEXDEFRAG" with "ALTER INDEX" with "REORGANIZE" option.

Re-write the code, replace "DBCC SHOWCONTIG" with "sys.dm_db_index_physical_stats".

Use of DBCC PINTABLE/DBCC UNPINTABLE is not required and has been removed to prevent additional problems. The syntax for this command still works but does not affect the server.

Refer to SQL Server books online for equivalent DMVs and DMFs that you may want to use instead of deprecated and discontinued DBCC commands.

More info

- Deprecated Database Engine Features in SQL Server
- Discontinued Database Engine Functionality in SQL Server

Impacted objects

Type	Name
Procedure	dbo.DbccNewAllocProcedure
Procedure	dbo.DbccRowLockProcedure
Procedure	dbo.DbccTextAllocProcedure
Procedure	dbo.DbccTextAllProcedure

Object details

Type: Procedure
Name: dbo.DbccNewAllocProcedure

The specific DBCC command is discontinued. For more details, please see: Line 5, Column 9.

Recommended Fix(s)

No Suggested Fix

Export report

10:39 PM 7/19/2018

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Walkthrough

Database Experimentation Assistant

Database Experimentation Assistant

Welcome to Database Experimentation Assistant

To get started, click on the left side navigation bar.

Capture a workload on the source server.

Replay the captured workload on target 1 and target 2.

Analyze replayed traces collected from target 1 and target 2.

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Database Experimentation Assistant

Connect to server to view existing reports or generate new reports

Server name

Report Server e.g. localhost

Authentication Type

Windows

Encrypt connection

Trust server certificate

Connect

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Database Experimentation Assistant

Analysis Reports

Connected Report Server: DEAGAWSVR2016 Switch Server

+ NEW REPORT

STATUS	NAME	DATE
✓	Test08To14Sma	2018/07/18
✓	ADWTPS100	2018/07/18
✓	2016Migration	2018/03/13

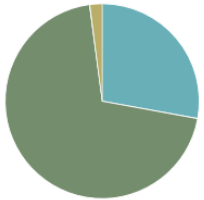
2016Migration

Threshold 5% Export Print

Target	Instance Name	Product Name	Trace File
Target 1	SQL2008SOURCE	SQL Server 2008 SP4 MS15-059	C:\Users\moljain\Documents\sql20008-small\Trace.trc
Target 2	SQL2016TRACE	SQL Server 2016	C:\Users\moljain\Documents\sql2016-small\Trace.trc

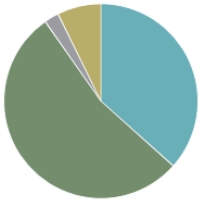
QUERY DISTRIBUTION

By execution count



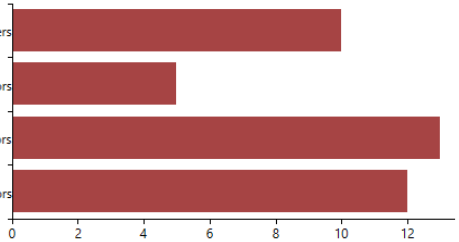
Degraded [14701] 27.9%
Improved [36951] 70%
Cannot Evaluate [1] 0%
Same [1108] 2.1%

By distinct queries



Degraded [15] 36.6%
Improved [22] 53.7%
Cannot Evaluate [1] 2.4%
Same [3] 7.3%

Distinct error queries count by type



Error ID count

Error ID 156 (11) 36.7%
Error ID 208 (6) 20%
Error ID 2714 (5) 16.7%
Error ID 3701 (5) 16.7%
Error ID 539 (3) 10%

Top queries in workload

Top Improved Queries | Top Degraded Queries

Hash Id	Query Text	Mean Duration on Target 1 (µs)	Mean Duration on Target 2 (µs)	Duration Difference (µs)
-1616155278339986948	IF OBJECT_ID ((STR), (STR)) IS NOT NULL DROP TABLE DBO.EMPLOYEEONE;	102353	9105	-93248
-1704013646769530258	SELECT DISTINCT PP.LASTNAME, PP.FIRSTNAME FROM PERSON.PERSON PP JOIN	181508	127802	-53706

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Database Experimentation Assistant

Analysis Reports

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+ NEW REPORT

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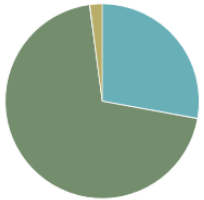
Target 2	SQL2016TRACE	SQL Server 2016	C:\Users\moljain\Documents\sql2016-small\Trace.trc

QUERY DISTRIBUTION

By execution count



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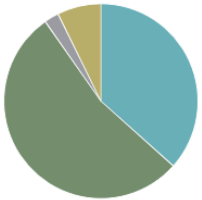

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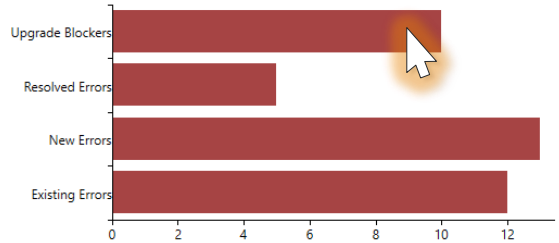

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Top queries in workload

Top Improved Queries | Top Degraded Queries

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Database Experimentation Assistant

Analysis Reports

Connected Report Server: DEAGAWSVR2016 Switch Server

+ NEW REPORT

STATUS	NAME	DATE
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2016Migration > Error

Threshold 5% Export Print

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ERROR QUERIES

Error Type	Query Count	Execution Count
Existing Errors (Errors on target 1 server that continue to exist on the target 2 server.)	12	1128
New Errors (Errors which are new on the target 2 server.)	13	3552
Resolved Errors (Errors which existed on target 1 server but resolved in target 2 server.)	5	14
Upgrade Blockers (Errors that will block upgrade to target server.)	10	3519

New Errors | Existing Errors | Resolved Errors | Upgrade Blockers

Query Text	Error ID	Error Count	Error Text
SELECT PRODUCTID, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, LINETOTAL COMPUTE SUM(LINETOTAL) BY PRODUCTID;	156	372	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, SUM(LINETOTAL) AS TOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} GROUP BY PRODUCTID, ORDERQTY ORDER BY PRODUCTID, ORDERQTY COMPUTE SUM(SUM(LINETOTAL)) BY PRODUCTID, ORDERQTY COMPUTE SUM(SUM(LINETOTAL));	156	397	Incorrect syntax near the keyword 'COMPUTE'.
SELECT SALESPERSONID, CUSTOMERID, ORDERDATE, SUBTOTAL, TOTALDUE FROM SALES.SALESORDERHEADER ORDER BY SALESPERSONID, ORDERDATE COMPUTE SUM(SUBTOTAL), SUM (TOTALDUE) BY SALESPERSONID;	156	342	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, LINETOTAL FROM SALES.SALESORDERDETAIL COMPUTE SUM (ORDERQTY), SUM(LINETOTAL);	156	365	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} COMPUTE SUM(ORDERQTY), SUM(LINETOTAL);	156	327	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, ORDERQTY, LINETOTAL COMPUTE SUM(LINETOTAL) BY PRODUCTID, ORDERQTY COMPUTE SUM(LINETOTAL) BY PRODUCTID;	156	354	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, LINETOTAL COMPUTE SUM(LINETOTAL), MAX(LINETOTAL) BY PRODUCTID;	156	375	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID COMPUTE SUM(ORDERQTY), SUM(LINETOTAL) BY PRODUCTID COMPUTE SUM(ORDERQTY), SUM(LINETOTAL);	156	323	Incorrect syntax near the keyword 'COMPUTE'.
SELECT CUSTOMERID, ORDERDATE, SUBTOTAL, TOTALDUE FROM SALES.SALESORDERHEADER WHERE	156	331	Incorrect syntax near the keyword 'COMPUTE'.

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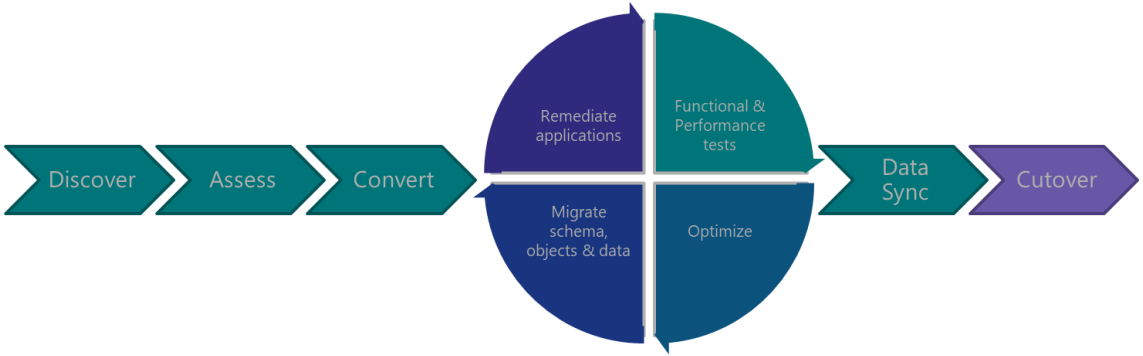
Post-Migration



I moved the data, am I done?

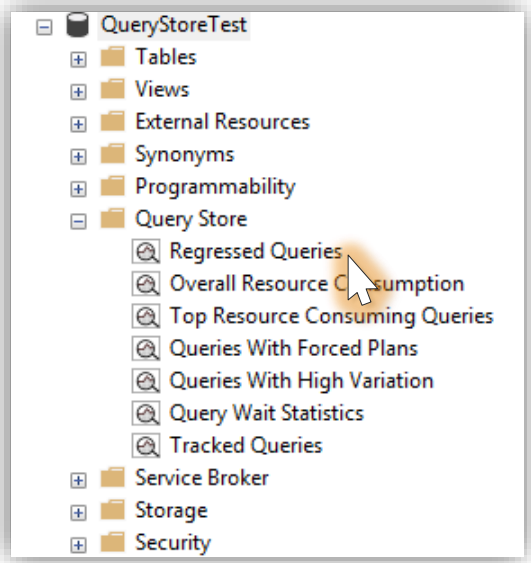
SQL Server post migration step is very crucial for reconciling any data accuracy and completeness, as well as uncover performance issues with the workload.

Recommended Upgrade Plan for latest DB Compatibility Level:

Upgrading DB Compat until recently

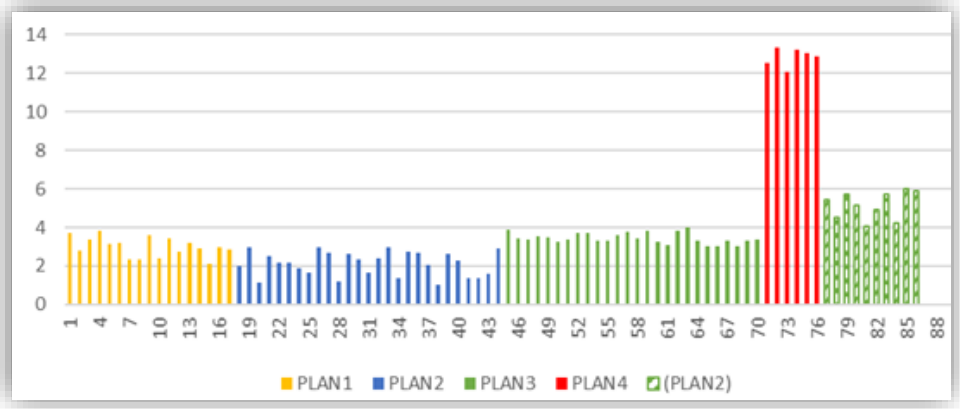
SQL 2016+

Query Store Regressed Queries



SQL 2017+

Automatic Plan Correction



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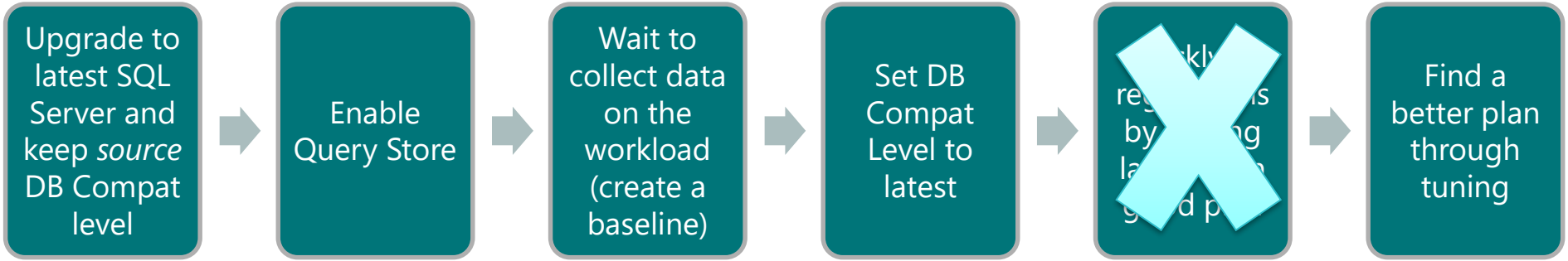
Query Tuning Assistant (QTA)

1st Priority: to guide users through the documented and recommended DB Compatibility upgrade procedure with ease.

What if instead of choosing between current and last know good plan, we find a 3rd, better plan?



Query Tuning Assistant (QTA) Workflow

Available in SSMS v18 and Powershell (preview)

Analysis and Recommendations

- Correlation vs Independence
  `ASSUME_MIN_SELECTIVITY_FOR_FILTER_ESTIMATES`
- Simple vs Base Join Containment
  `ASSUME_JOIN_PREDICATE_DEPENDS_ON_FILTERS`
- TVF fixed estimation of 100 rows vs 1 row
  'QUERYTRACEON 9488'
- As a last resort, full-fledge use of CE70 is also considered, when all else doesn't yield desired results.

Modernization Tools Breakdown

DMA

- Readiness assessment: blocking issues breaking changes, behavior changes
- Moves schema, data and uncontained objects (like logins) To Azure SQL Database
- Backup / Restore to another SQL Server (Keeps source DB Compatibility Level)
- New feature recommendation

DEA

- A/B Testing
- Capture and Replay workload for performance testing and reporting
- Also reports on migration blockers because of failed T-SQL syntax

QTA

- Upgrade Database Compatibility Model to desired state
- Detects workload regressions, and tests CE model variations (subsets)
- Provides tangible recommendations for tuning queries without reverting DB compat

Session takeaways

- Start planning for End of Service for SQL Server 2008/R2 and Windows Server 2008/R2 today!
- Review the database migration guide
- Familiarize yourself with the DMA, DEA, Query Store and Query Tuning Assistant
- Leverage Database Compatibility to accelerate modernization

Session resources

http://aka.ms/sqleosfaq - End-of-Support FAQ

Upgrade SQL Server

Database Migration Guide

Microsoft Assessment and Planning Toolkit

Overview of Data Migration Assistant

DEA 2.1 General Availability: Release Overview – Database Experimentation Assistant

Post-migration Validation and Optimization Guide

http://aka.ms/dbcompat (DB Compatibility Level based upgrades)

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Questions?

Thank You